

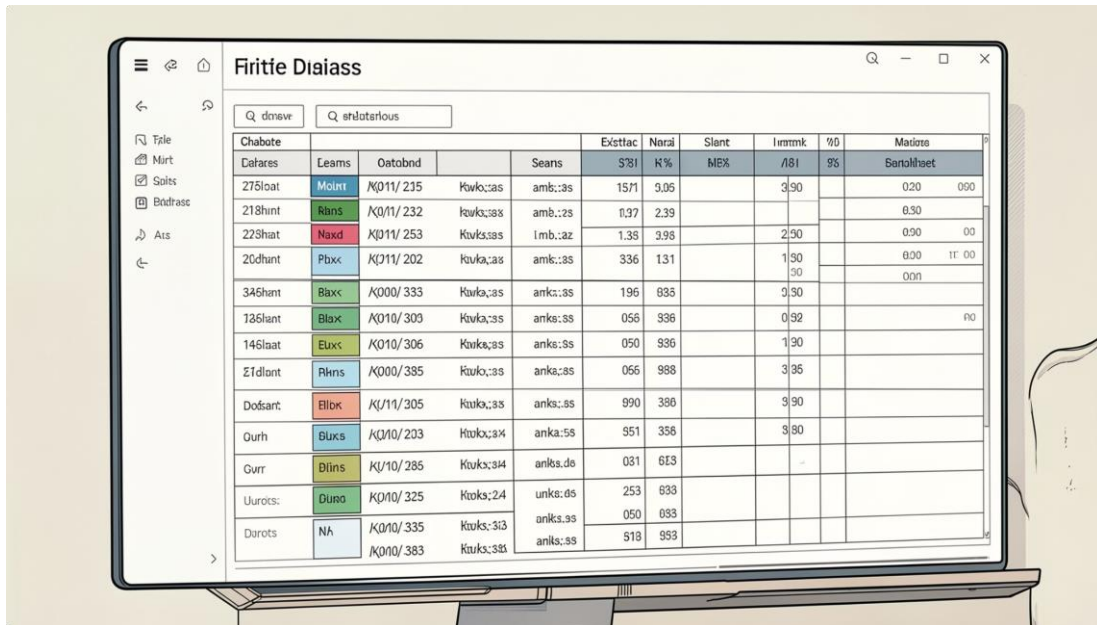
Job Market Analysis Project

Insights from **742 job postings** sourced from Sample Dataset — covering salary trends, geographic distribution, skill demand, industry hiring patterns, and education-compensation relationships.

PYTHON · EXCEL · MYSQL · POWER BI



Dataset at a Glance



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275loat	Molint	K011/ 215	Kowk:as	amb:as	1571	9.05		2.90		0.20	000
218hint	Rans	K011/ 232	Kowk:as	amb:as	1197	2.39				0.50	
223hat	Naxd	K011/ 253	Kowk:as	lmb:az	1.35	3.98		2.90		0.90	00
20dhint	Pbxc	K011/ 202	Kowk:as	amb:as	336	1.31		1.90		0.00	11 00
346hint	Baxc	K000/ 333	Kowk:as	ank:as	196	0.35		9.90			
126lant	Blax	K010/ 309	Kowk:as	ank:as	056	9.36		0.92			00
146laot	Elux	K010/ 306	Kowk:as	ank:as	050	9.36		1.90			
21dlaot	Rlms	K000/ 385	Kowk:as	anka:as	066	9.88		3.35			
Dofant	Elor	K011/ 305	Kowk:as	ank:as	990	3.96		9.90			
Ourh	Blux	K010/ 203	Kowk:as	anka:as	551	3.56		3.80			
Gurr	Blms	K010/ 285	Kowk:as	ank:as	031	6.13					
Uurots	Duno	K010/ 325	Kowk:as	ank:as	253	6.33					
Darots	NA	K010/ 335	Kowk:as	ank:as	050	0.33					
		K010/ 383	Kowk:as	ank:as	518	8.53					

742 Rows · 42 Columns

Source: Sample Dataset · Table:
job_market_analysis (MySQL)

Key Features

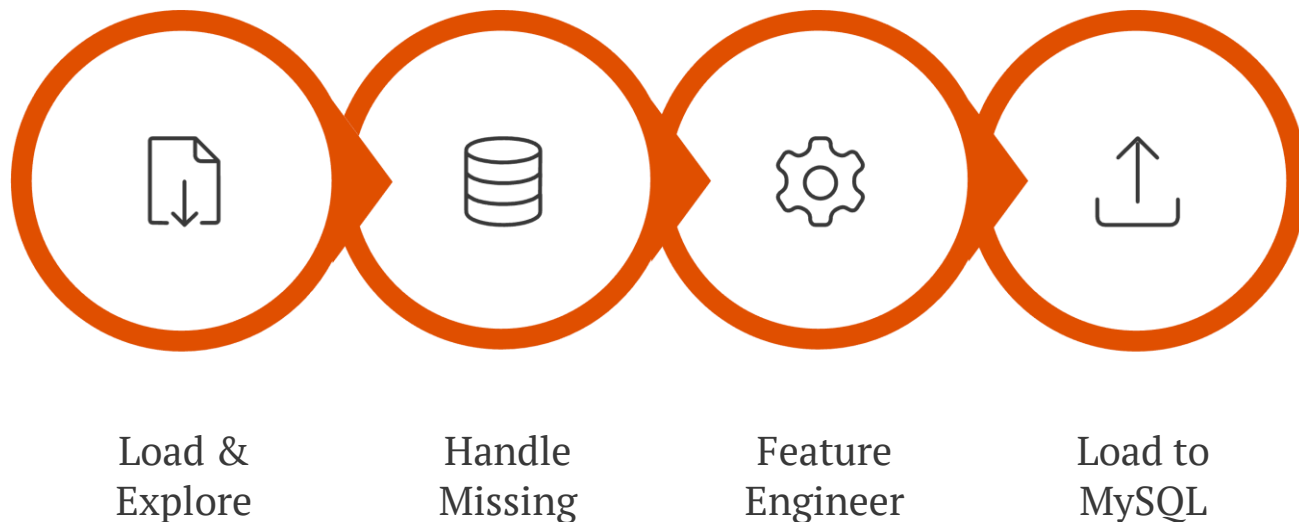
Job details, company info, location, 10
binary skill columns, education level

Data Quality

Missing values (-1 / '-1') handled during
cleaning; FIELD4 dropped (empty)

Data Cleaning & Preparation

Python (Pandas) preprocessing before loading into MySQL for analysis.



Key steps included replacing -1 values with column medians, extracting state codes from location strings, removing appended ratings from company names, and resetting sequential IDs from 0–741.



Geographic Distribution: Where the Jobs Are

152

California Jobs

#1 state — 20.5% of all postings

103

Massachusetts Jobs

#2 state — driven by biotech

44%

Top 3 Share

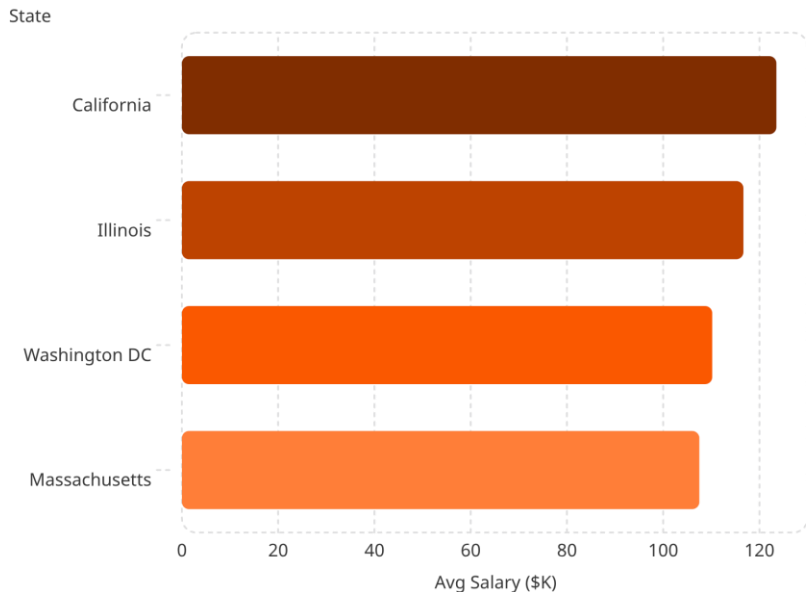
CA + MA + NY = 327 jobs

\$123K

CA Avg Salary

Highest state average nationally

Salary by State



National Average: **\$97K**

California and Illinois are significantly above the national average. CA also offers the widest salary range — **\$92.43K min to \$154.60K max** — the strongest negotiation potential of any state.

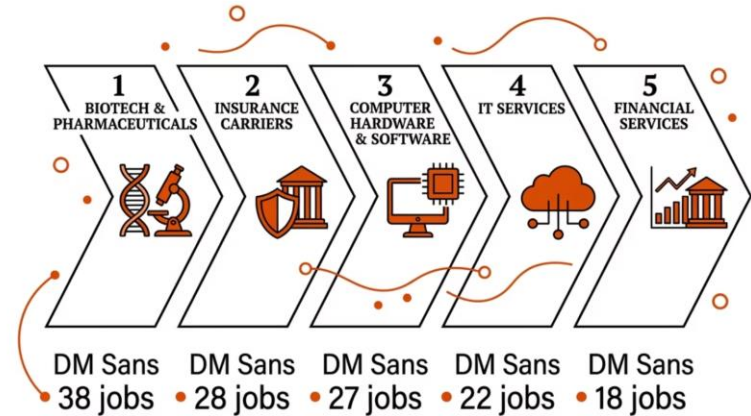
i Illinois (\$116.66K) and DC (\$110.18K) offer strong salaries with lower competition than CA/NY.

Top Industries & Employers

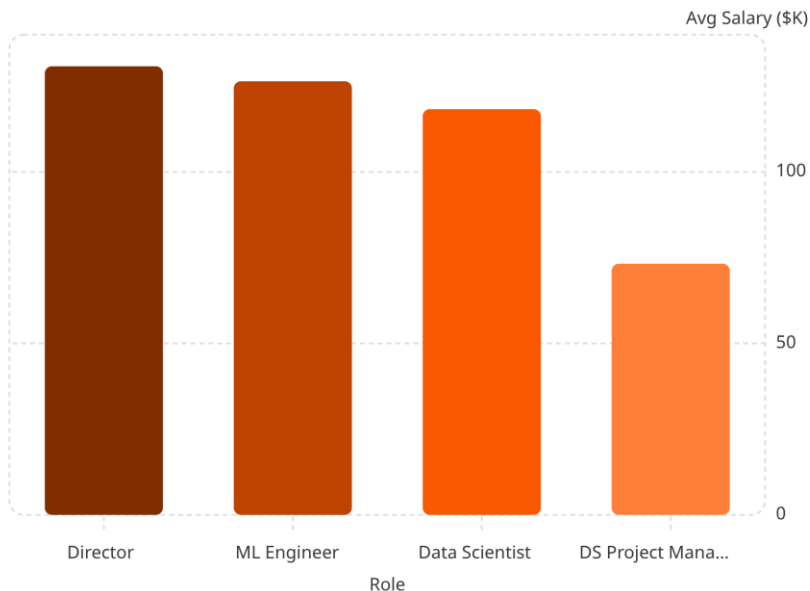
Surprising Finding

Biotech & Pharmaceuticals leads all industries with 38 jobs — not IT as commonly assumed. Insurance Carriers (28) and Computer Hardware & Software (27) rank 2nd and 3rd.

Top employers: **MassMutual**, **Takeda Pharmaceuticals**, and **Reynolds American** — each with 14 openings. Insurance and pharma companies dominate hiring, not tech giants.



Job Titles & Salary Hierarchy



A \$57.58K Salary Gap

Director-level roles earn the most at \$130.8K. ML Engineers command a significant premium over general data roles at \$126.43K.

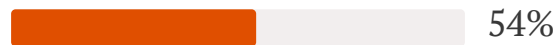
Data Scientist is the most common title — 131 postings, 2.5× more than Data Engineer (53). Combined DS variants total **173 postings (23.3% of all jobs)**.

Skills in Demand



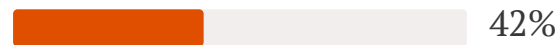
Python

516 jobs — universal foundation



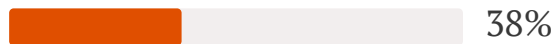
SQL

402 jobs — required across all roles



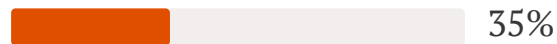
AWS

310 jobs — now a standard requirement



Spark / PySpark

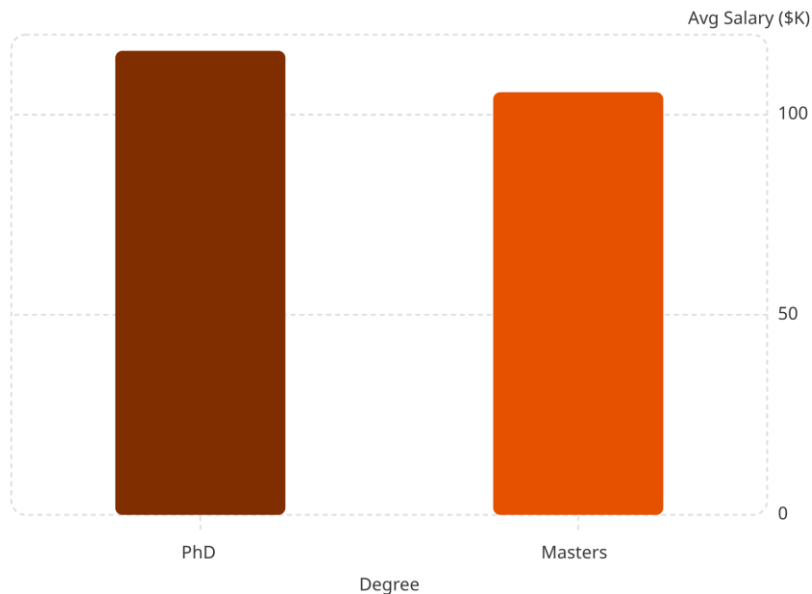
285 jobs — high demand for Data Engineers



Excel

260 jobs — widely required for analyst roles

Education & Salary



PhD Premium: 9.8%

PhD holders earn \$115.98K vs. Masters at \$105.63K — but Masters holders have **2.4× more job opportunities** (252 vs. 107).

- ✓ **51.6% of all jobs (383 postings) require no specific degree** — skills-based hiring is the dominant trend in Data Science.

Key Findings & Recommendations



Target California

#1 in jobs (152) and salary (\$123.51K). MA and IL are strong alternatives.



Master Python First

Required in 70% of jobs. Add SQL (54%) and Spark (38%) next.



Look Beyond Tech Giants

Biotech & Pharma leads hiring. MassMutual, Takeda, Reynolds American top employer list.



Skills Over Degrees

51.6% of jobs require no degree. Masters offers best job-to-salary balance.